

Gaussian Splatting and Neural Rendering for Dynamic 3D Scene Reconstruction: A Survey

Gaussian splatting and neural rendering have rapidly changed dynamic 3D scene reconstruction. Neural radiance fields introduced differentiable volumetric scene representations, while 3D Gaussian splatting made real-time novel-view synthesis and explicit point-based optimization practical at high quality. Dynamic and 4D settings extend these ideas to motion, deformation, non-rigid objects, SLAM, avatars, driving scenes, and live capture. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around Gaussian representations, dynamic and deformable modeling, radiance-field foundations, scene reconstruction and mapping, rendering efficiency, geometry estimation, human avatars, datasets, and applications. The survey argues that dynamic reconstruction is a joint problem of representation, camera pose, temporal correspondence, geometry regularization, and rendering speed. We close with open challenges in long-term dynamics, topology changes, physical consistency, compression, real-time capture, and evaluation.

Keywords:

3D Gaussian splatting; neural rendering; dynamic scene reconstruction; neural radiance fields; novel view synthesis; SLAM; 4D reconstruction

1 Introduction

Dynamic 3D scene reconstruction aims to recover geometry, appearance, camera motion, and temporal change from images or videos. Neural rendering made this problem differentiable by representing scenes with neural radiance fields, implicit functions, or learned features. 3D Gaussian splatting (3DGS) then shifted the field toward explicit, optimizable, point-based primitives that can be rasterized efficiently while preserving photorealistic view synthesis [21, 47, 52, 55, 65, 131, 227, 286]. The result is a rapidly expanding literature that connects computer vision, graphics, SLAM, robotics, avatars, and simulation.

Dynamic scenes make reconstruction harder than static novel-view synthesis. A method must handle moving objects, camera motion, occlusions, changing lighting, non-rigid deformation, topology changes, and limited observations. Gaussian splatting helps because Gaussians can move, split, merge, deform, and render quickly; neural rendering helps because

learned features and radiance fields can regularize sparse or ambiguous observations [21, 47, 52, 55, 65, 131, 227, 286]. The central design question is how to represent time: explicit per-frame primitives, canonical spaces with deformation fields, scene flow, factorized bases, or learned temporal priors.

This survey reviews Gaussian splatting and neural rendering for dynamic 3D scene reconstruction. We focus on representations, optimization, camera tracking, temporal modeling, geometry recovery, rendering speed, human avatars, datasets, and evaluation. The central claim is that dynamic reconstruction is not just a better renderer; it is a systems problem coupling representation, correspondence, optimization, and measurement.

2 Background

2.1 Neural Rendering and Radiance Fields

Neural radiance fields represent a scene as a continuous function mapping position and direction to density and color. They established a powerful differentiable rendering framework for novel-view synthesis, but early methods were slow and often optimized per scene. Subsequent work improved encoding, sampling, acceleration, generalization, and dynamic extensions. Radiance-field methods remain important because they provide a conceptual foundation for differentiable volumetric reconstruction [21, 47, 52, 55, 65, 131, 227, 286].

2.2 3D Gaussian Splatting

3DGS represents scenes as anisotropic Gaussians with position, covariance, opacity, and appearance parameters. Instead of sampling a neural field along rays, it rasterizes splats efficiently. This explicit representation enables real-time rendering, direct optimization, and interpretable primitive manipulation. It also raises new questions about densification, regularization, pruning, compression, and geometric fidelity [21, 47, 52, 55, 65, 131, 227, 286].

2.3 Dynamic Scene Reconstruction

Dynamic reconstruction extends static view synthesis by adding time. The scene may be decomposed into a static background and moving foreground, modeled in a canonical space with deformation fields, represented by per-time primitives, or tracked through scene flow. The choice affects memory, temporal consistency, editability, and real-time performance [21, 47, 52, 65, 95, 131, 227, 286].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with Gaussian splatting, 3DGS, dynamic Gaussian splatting, 4D Gaussian splatting, deformable Gaussians, neural rendering, dynamic NeRF, neural radiance fields, novel view synthesis, dynamic 3D scene reconstruction, Gaussian SLAM, neural SLAM, radiance-field SLAM, surface reconstruction, depth, camera pose, multi-view reconstruction, human avatars, autonomous driving, real-time rendering, scene flow, and point-based neural rendering. Records were retained when title or

abstract evidence connected Gaussian splatting, neural rendering, radiance fields, or related point/implicit representations with dynamic scenes, 3D reconstruction, view synthesis, mapping, geometry, humans, evaluation, or applications. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize the literature along six axes. The first is *representation*: radiance field, voxel grid, point cloud, anisotropic Gaussian, mesh, surfel, hybrid explicit-implicit model, or learned feature field. The second is *time model*: independent frames, deformation field, canonical space, scene flow, trajectory basis, motion graph, or streaming update. The third is *input*: calibrated images, monocular video, RGB-D, multi-view video, LiDAR, event cameras, or SLAM streams. The fourth is *optimization*: photometric loss, depth regularization, geometry priors, tracking, bundle adjustment, densification, pruning, and temporal consistency. The fifth is *output*: novel views, geometry, camera trajectory, dynamic object motion, avatar animation, or simulation asset. The sixth is *evaluation*: rendering fidelity, geometry accuracy, temporal stability, speed, memory, robustness, and downstream utility.

3.3 Gaussian Representations and Optimization

Gaussian splatting methods optimize explicit primitives. Position and covariance encode geometry; opacity and appearance encode visibility and color; densification and pruning adapt model capacity. Dynamic methods add temporal trajectories, deformation fields, or per-frame updates. Optimization must avoid floating artifacts, over-densification, temporal jitter, and poor geometry while retaining fast rendering [21, 47, 52, 55, 65, 131, 227, 286].

3.4 Dynamic and Deformable Modeling

Dynamic 3DGS and dynamic NeRF methods must model non-rigid motion and occlusion. Canonical-space methods learn a stable template and map observations into it; per-frame methods optimize separate primitives and regularize across time; scene-flow methods estimate motion directly. Each strategy trades memory, robustness, and ability to handle topology change [21, 47, 52, 65, 95, 131, 227, 286].

3.5 Scene Reconstruction, Mapping, and SLAM

SLAM-oriented neural rendering uses images or RGB-D streams to estimate camera pose and incrementally build a scene representation. Gaussian SLAM is attractive because splats can be added and rendered quickly, but online settings expose tracking drift, loop closure, map growth, and dynamic-object contamination. Robust systems combine photometric alignment, depth cues, geometry priors, and map maintenance [13, 37, 41, 52, 55, 65, 109, 131].

3.6 Geometry, Depth, and Surfaces

Photorealistic rendering does not guarantee accurate geometry. Methods therefore add depth supervision, normal regularization, surface extraction, multi-view consistency, or mesh conversion. Dynamic scenes make geometry harder because motion can explain away depth errors. Evaluation should report both rendering metrics and geometry metrics [21, 47, 52, 55, 65, 95, 109, 131].

3.7 Rendering Efficiency and Compression

One reason for the popularity of 3DGS is speed. Real-time rendering enables interactive capture, VR/AR, robotics, and simulation. However, dense Gaussian sets can consume substantial memory. Compression, pruning, quantization, level of detail, and efficient rasterization are therefore central for deployment [21, 47, 52, 55, 65, 131, 227, 286].

3.8 Humans, Avatars, and Articulated Scenes

Humans are a major application area because bodies and faces are dynamic, articulated, and visually important. Avatar methods combine body priors, skeletons, canonical spaces, expression models, and Gaussian or neural rendering. The challenge is preserving identity and clothing details while supporting novel poses and viewpoints [31, 95, 197, 209, 233, 261, 264, 298].

3.9 Datasets, Evaluation, and Applications

Applications include telepresence, robotics, autonomous driving, cultural heritage, medical visualization, film production, and simulation. Datasets vary in camera setup, calibration, motion, lighting, and ground truth. Evaluation should separate

static fidelity, dynamic consistency, geometry accuracy, and runtime. For dynamic scenes, temporal stability and behavior under occlusion are as important as PSNR or LPIPS [13, 21, 47, 52, 55, 65, 227, 286].

3.10 Representative Literature Map

The following map cites the full retained corpus by theme, making the survey coverage explicit.

3d gaussian splatting representations and optimization The 3d gaussian splatting representations and optimization cluster contains work that develops methods, systems, datasets, or analyses central to this survey [21, 47, 52, 55, 65, 95, 109, 131, 143]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [37, 91, 106, 177, 201, 233, 241, 243, 281]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [62, 92, 163, 166, 172, 209, 230, 269, 272]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [15, 75, 82, 155, 207, 231, 251, 257, 278]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [31, 69, 90, 98, 111, 189, 240, 275, 289]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [78, 102, 128, 190, 238, 245, 264, 295, 298]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [117, 158, 186, 211, 261, 265, 268, 279, 280]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [57, 86, 93, 118, 126, 176, 235, 271, 300]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [4, 26, 48, 74, 119, 144, 212, 219, 270]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [19, 38, 54, 81, 88, 159, 161, 168, 196]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 8, 44, 45, 96, 149, 175, 246, 282]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [33, 99, 120, 129, 145, 173, 182, 185, 214]. Additional retained studies broaden the cluster with alter-

native architectures, training signals, data sources, and deployment assumptions [22, 46, 50, 61, 94, 105, 183, 202, 213]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [39, 60, 64, 127, 150, 157, 181, 237, 247]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [14, 40, 79, 174, 179, 232, 234, 274, 291]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [10, 16, 56, 112, 208, 217, 252, 266, 293]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [5, 7, 43, 130, 135, 215, 220, 239, 250]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [17, 23, 25, 32, 71, 73, 76, 100, 103]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [1, 49, 72, 154, 205, 218, 255, 273, 296]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [24, 27, 53, 77, 113, 122, 138, 141, 142]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [51, 83, 107, 132, 136, 164, 184, 256, 292]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [18, 80, 87, 101, 110, 151, 203, 260, 285]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [35, 59, 66, 133, 170, 225, 254, 287, 290]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [9, 29, 84, 171, 180, 199, 210, 259, 283]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 28, 85, 108, 140, 191, 224, 228, 229]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [89, 97, 134, 148, 153, 165, 258, 277, 297]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [12, 30, 63, 115, 206, 222, 223, 248, 263]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [42, 147, 178, 187, 188, 226, 244, 284, 294]. Addi-

tional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [34, 58, 68, 123, 146, 152, 242, 253, 262]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [162, 193, 200, 204].

Dynamic, deformable, and 4d scene modeling The dynamic, deformable, and 4d scene modeling cluster contains work that develops methods, systems, datasets, or analyses central to this survey [36, 67, 70, 114, 124, 169, 192, 227, 286]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [2, 116, 236].

Neural radiance fields and neural rendering foundations The neural radiance fields and neural rendering foundations cluster contains work that develops methods, systems, datasets, or analyses central to this survey [13, 41, 104, 125, 267, 288].

Scene reconstruction, mapping, and slam The scene reconstruction, mapping, and slam cluster contains work that develops methods, systems, datasets, or analyses central to this survey [20, 156, 198, 249].

View synthesis, rendering efficiency, and compression The view synthesis, rendering efficiency, and compression cluster contains work that develops methods, systems, datasets, or analyses central to this survey [121, 216].

Geometry, depth, surfaces, and camera estimation The geometry, depth, surfaces, and camera estimation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [11, 137, 139, 167, 197].

Datasets, evaluation, and applications The datasets, evaluation, and applications cluster contains work that develops methods, systems, datasets, or analyses central to this survey [160, 194, 195, 221, 276, 299].

4 Challenges and Prospects

4.1 Temporal Consistency and Topology Change

Dynamic methods often assume smooth motion, but real scenes include contacts, disocclusions, topology

changes, and appearance changes. Future representations need to support long sequences without accumulating temporal artifacts or exploding memory.

4.2 Geometry Versus Appearance

Photometric metrics reward good images, not necessarily good 3D structure. Gaussian splatting can produce excellent views with imperfect geometry. Progress requires geometry-aware losses, calibrated datasets, and evaluation protocols that report surface, depth, and camera accuracy.

4.3 Online Reconstruction and SLAM Robustness

Real-time reconstruction must handle pose drift, moving objects, exposure changes, and limited compute. Gaussian maps make online rendering fast, but long-term map management and loop closure remain difficult.

4.4 Compression, Editing, and Simulation

Deployment requires compact, editable, and physically meaningful scene assets. Future work will likely combine Gaussian rendering with meshes, physics priors, semantic labels, and compression so reconstructed scenes can be reused beyond view synthesis.

5 Conclusion

Gaussian splatting and neural rendering have reshaped dynamic 3D reconstruction by combining differentiable view synthesis with explicit, efficient scene representations. This survey reviewed 3DGS, dynamic and deformable modeling, radiance-field foundations, SLAM, geometry, efficiency, human avatars, datasets, and applications. The central lesson is that dynamic reconstruction requires jointly solving appearance, geometry, pose, and time. Robust future systems will need temporally stable representations, accurate geometry, online mapping, compact storage, and evaluation beyond static rendering metrics.

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